OpenVLA-OFT · arXiv 2502.19645 · Kim, Finn, Liang (Stanford)

Scientific Reviewer

8+3

Not superseding FM · L1 ≈ diffusion cheaper

Caveats: LoRA vs full-FT π_0 · ms vs seconds

Oct 26 paper club · VLA Architectures

SPINE: AR discrete bins → parallel continuous L1 chunks

OFT ≠ FAST ≠ FM · FT recipe / speed+success patch · not $\pi_{0.5}$ on BEHAVIOR

TAXONOMY · OFT ≠ FAST ≠ FM

Three parallel improvement paths — do not collapse names

OpenVLA-OFT

FT recipe

Parallel continuous
L1 chunks
K=8 / 25

Decode: 1-pass
Loss: L1
Still 7B OpenVLA

FAST

Tokenizer

DCT→BPE discrete
chunk tokens
still AR

Decode: serial AR
Loss: CE
~750 ms/chunk class

π_0 / $\pi_{0.5}$

FM deploy

Continuous FM
chunks / NFE
OpenPI stack

Decode: denoise
Loss: FM / ε
π_0 ALOHA ~291 Hz

Vanilla OpenVLA = AR discrete bins @ ~4.2 Hz → OFT rewrites output signature

OFT does NOT replace $\pi_{0.5}$ for 2026 BEHAVIOR · OFT ≠ FAST tokenizer swap

REVIEWER · Strengths

Factorial study	decode × representation × objective on one base VLA
Throughput + latency	exact tables; diffusion NFE quality/compute tie
Dual venue	LIBERO standardized + hard real ALOHA shift
FiLM ablation	language → chance without it — honest
L1 ≈ diffusion	actionable Empiricist result; cheaper stack
Open release	code + HF checkpoints

REVIEWER · Caveats

LoRA OFT vs full-FT π_0	train-cost not equated; PyTorch vs JAX
97.1% needs wrist+proprio+filtered	fair only within that block
Put-in-pot ~50%	“+15%” relative, not product-ready
L1 multimodal undertested	Sec VIII admits median-mode limit
Abstract ms vs table seconds	cite Table II/III (~0.073 s / 0.321 s)
OFT for pretrain open	studied for FT only

OUTPUT SIGNATURE SHIFT

Vanilla OpenVLA → OpenVLA-OFT

Axis	Vanilla OpenVLA	OFT
Decode	Serial AR discrete bins	Parallel continuous decode
Chunk	K=1	K=8 LIBERO / K=25 ALOHA
Head	256-bin CE	Continuous MLP + L1
Hz	~4.2	~109.7 LIBERO (26×); ALOHA ~77.9 (43×)
LIBERO SR	76.5%	97.1% (π_0 94.2%)

OFT+ adds FiLM on SigLIP+DINOv2 (multi-view language) — without FiLM, language → ~chance

CITE OFT FOR / DO NOT CITE FOR

Cite for

- ✓ FT recipe: parallel + chunk + Cont + L1
- ✓ Output signature shift vs vanilla
- ✓ High-Hz FT of large VLMs w/o LL controller
- ✓ Fair ablation morals vs Cont-Diff / PD&AC
- ✓ L1 ≈ diffusion SR at lower cost

Do not cite for

- ✗ Universal replacement for FM heads
- ✗ BEHAVIOR q-score SOTA
- ✗ Proof L1 models multimodal demos
- ✗ Fair ALOHA win without n/rubric/JAX notes
- ✗ Drop-in household champion

Fig. 2: Key design decisions for VLA fine-tuning. Left: Comparison between autoregressive decoding, which generates actions sequentially, and parallel decoding, which leverages bidirectional attention and generates all actions in a single forward pass. **Right**: Comparison between discrete action tokens with next-token prediction and continuous action values with L1 regression or diffusion modeling objectives. The original OpenVLA training scheme includes autoregressive decoding, discrete actions, and next-token prediction.

ing. While these diffusion-based VLAs achieve higher action throughput than autoregressive VLAs by generating multi-timestep action chunks simultaneously, they introduce computational trade-offs through slower training and multiple denoising or integration steps at inference time. Furthermore, these diffusion VLAs vary considerably in architecture, learning algorithm, vision-language fusion approach, and input-output specifications—and which design elements most significantly impact performance remains unclear. Through controlled experiments, we show that policies fine-tuned with a simpler L1 regression objective can match more complex approaches in

timestep action takes 0.33 seconds on an NVIDIA A100 GPU. For a chunk size of K timesteps and action dimensionality D, OpenVLA requires KD sequential decoder forward passes versus just D passes without chunking. This K-fold increase in latency makes action chunking impractical for high-frequency robots under the original formulation. In the next section, we present a parallel generation scheme that enables efficient action chunking.

IV. Studying Key VLA Fine-Tuning Design Decisions

LIBERO Table I · SR detail

Method	Spatial	Object	Goal	Long	Avg
OpenVLA FT (vanilla)	84.7	88.4	79.2	53.7	76.5
+ PD&AC discrete	91.3	92.7	90.5	86.5	90.2
+ PD&AC Cont-Diff	96.9	98.1	95.5	91.1	95.4
OFT Cont-L1 (3rd)	96.2	98.3	96.2	90.7	95.3
OFT +wrist+proprio	**97.6**	**98.4**	**97.9**	**94.5**	**97.1**
π_0 FT	96.8	98.8	95.8	85.2	94.2

PD+AC ≈ +13.7 pp · continuous ≈ +5 pp · Long cell is the interesting win (94.5 vs π_0 85.2)

SCIENTIFIC VERDICT

Strong systems / recipe paper.

Parallel continuous L1 chunks = better OpenVLA FT default

than AR discrete CE — for speed and success.

Does NOT license

“OFT supersedes FM.”

Cite for FT recipe / signature shift — not BEHAVIOR q-score SOTA

Scientific Reviewer · close

Better OpenVLA FT default — does NOT license "OFT supersedes FM."

Cite parallel continuous L1 chunks · Caveat LoRA vs full-FT · prefer Table II/III seconds

Q&A · Oct 26 · OpenVLA-OFT / 2502.19645

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